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Content versus Label Claims in Cannabidiol (CBD)-Containing Products Obtained from Commercial Outlets in the State of Mississippi

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ABSTRACT

Products containing cannabidiol (CBD) are now available throughout the United States, but their quality is oftentimes questionable. The CBD and Δ^9-tetrahydrocannabinol (THC) content of 25 commercially available hemp oil products, obtained throughout the state of Mississippi, was determined *via* gas chromatography/flame ionization detection (GC/FID). These products were also analyzed for the presence of synthetic cannabinoids using full scan gas chromatography/mass spectrometry (GC/MS). Analytical findings were compared to label claims for CBD content. Product label claims for CBD ranged from no claim to 500 mg per serving; however, marked variability was observed between actual CBD content and claimed quantities. Of the 25 products, only three were within ±20% of label claim. Fifteen were well below the stated claim for CBD; two exceed claims in excess of 50%; and 5 made no claims. In addition, THC content for three products exceeded the 0.3% legal limit. Furthermore, four products—primarily marketed for vaping—were adulterated with synthetic cannabinoids. From this small, but diverse, sampling of hemp-derived merchandise, it appears that most product label claims do not accurately reflect actual CBD content and are fraudulent in that regard. Moreover, products that exceed legal THC levels may jeopardize a consumer's employment status (i.e. failed "drug test"), while those adulterated with synthetic cannabinoids may subject them to serious adverse health effects. These findings argue strongly for further development of current good manufacturing practices for CBD-containing products and their stringent enforcement.

KEYWORDS

cannabidiol; CBD; phytocannabinoids; quality control; synthetic cannabinoids; Δ^9-tetrahydrocannabinol; THC

Introduction

The medicinal properties of *Cannabis sativa* (aka, marijuana) have been long been touted by scientists, clinicians and laypersons alike. *C. sativa's* pharmacological activity can be traced to a diverse set of phytochemicals, particularly the phytocannabinoids, present in the plant's flowering parts. Among those most recognized phytocannabinoids

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are Δ^9-tetrahydrocannbinol (THC), the principal psychoactive agent in *C. sativa*, and cannabidiol (CBD), a non-psychoactive compound linked to a variety of beneficial health effects (ElSohly et al. 2017). For many decades, research into the medical benefits of CBD was hampered by its close association with THC. This stemmed from the regulation of *C. sativa,* along with purified phytocannabinoid preparations, as Schedule I controlled substances. Over the last decade, however, many states have made "medical marijuana" available to consumers *via* prescription from a qualified medical practitioner. Rapid growth of the "medical marijuana" industry, along with its attendant dispensaries and growing public interest also brought additional focus onto the health benefits of the non-psychotropic cannabinoid, CBD. In June 2018, a purified oral solution of CBD (Epidiolex®) was approved by the US Food and Drug Administration (FDA) for treatment of certain drug-resistant seizures (Corroon and Kight 2018). Soon after, the US Congress passed the *Agriculture Improvement Act of 2018* (aka, the "Farm Bill"), which defined as "hemp" *C. sativa* variants containing <0.3% THC by dry weight, if grown under an approved state program, thus removing hemp and derived products, from the US Drug Enforcement Administration (DEA) list of controlled substances (H.R.2, 2018).

Passage of the "Farm Bill" not only unleashed a plethora of "hemp-derived" CBD products onto the US market, but it left their regulatory status unresolved. At present, except for Epidiolex®, CBD products are not drugs; nor are they legal as dietary supplements, although several stakeholder groups are advocating the FDA categorize them as dietary supplements. Consequently, the regulatory situation for CBD products on the US market is still murky, and in this environment, products have flooded the market, appearing in 'CBD stores', vape shops, convenience outlets, gas stations, along with pharmacies and health food stores.

The purpose of this study was to gather a sampling of hemp-derived products from across the state of Mississippi, analyze them for CBD content, and compare the results to product label claims. Our concern about truth in labeling of CBD products stemmed, in part, from earlier surveys conducted by this group regarding the quality of various dietary supplements (Gurley et al. 2000; Glisson and Walker 2010). Discrepancies in product label claims and actual phytochemical content are not uncommon for dietary supplements (Glisson and Walker 2010). However, despite its appearance in a wide variety of consumer products, CBD is not considered a dietary supplement by the FDA; its regulatory status is currently under deliberation. Nevertheless, given the novelty of CBD products and their current lack of a well-defined regulatory status, we deemed it necessary to conduct a preliminary survey of CBD product quality in our state. Our research team at the University of Mississippi and its National Center for Natural Products Research (NCNPR), along with ElSohly Laboratories, Inc. (ELI), have been at the forefront of *Cannabis* research for more than 50 years and is experienced in analyzing phytocannabinoids from a variety of biological and non-biological matrices (ElSohly and Slade 2005; ElSohly et al. 2017).

Materials and methods

Owing to the necessity to adhere strictly to DEA regulations and guidance at the time of this study, receipt of any *Cannabis* samples could only come from DEA registrants

or law enforcement. Therefore, officers of the Mississippi Bureau of Narcotics were tasked with purchasing CBD-containing products from local retailers (e.g. convenient stores, vape shops, CBD shops, etc.) across the state of Mississippi. Twenty-five oils and vaping products claiming to contain CBD were included in the analysis. Results from this survey would serve as an indicator for more rigorous sampling in the future, if needed. Products were brought to ELI, accessioned, and aliquots/sampleswere subjected to analysis for phytocannabinoids using GC/FID and using full scan GC/MS for synthetic cannabinoids analysis.

GC/FID analysis for THC and CBD

The GC/FID method used for the analysis of the products tested and reported in this manuscript is the same method previously reported by Ibrahim et al. (2018). The method was validated not only for THC and CBD but also for their acid precursors and other cannabinoids.

Briefly, solutions of the products in question are prepared in an organic solvent (hexane) at a specific concentration. Aliquots of the product's solutions are spiked with an internal standard (4-androstene-3,17-dione), the solvent evaporated and the residue derivatized using BSTFA in the presence of a catalyst (Dimethylaminopyridine, DMAP) to make the TMS derivatives of the cannabinoids. The derivatives were then analyzed by the GC/FID under the previously reported method (Ibrahim et al. 2018).

The limit of detection (LOD) was 0.12 µg/mL for CBD and 0.15 µg/mL for THC, while the limit of quantitation (LOQ) for CBD was 0.35 µg/mL and for THC was 0.45 µg/mL. Linearity was up to 100 µg/mL for both cannabinoids. The accuracy and precision of the method were within ± 15%.

GC/MS analysis for the synthetic cannabinoids

GC/MS analysis for the presence of synthetic cannabinoids was carried out on the underivatized solutions of the tested products in the full scan mode. The analysis was performed on an Agilent Technologies 7890 A gas chromatograph with an Agilent Technologies 5975 C MSD and an Agilent Technologies 7693 autosampler. Separation was achieved on an Agilent Technologies 25 M x 0.200 mm DB-5MS column (0.33 μ film). Helium was used as the carrier gas at a flow rate of 1.0 mL/min. The inlet was configured in splitless mode at a temperature of 250˚C. The temperature program started at 170˚C for one minute and then ramped up at 30˚C/minute to 300˚C for 14.67 min. The total run time was approximately 20 min. Data acquisition was performed on ChemStation G1701EA.

The qualitative identification of the synthetic cannabinoids in the tested samples was carried out based on comparison of the mass spectra of the peaks detected in the product(s) and the reported spectra (Cayman Chemical, Ann Arbor, MI) or based on direct comparison with a reference standard (R_t and spectral fragmentation).

Results

CBD product label claim, actual CBD content, percent of label claim, THC content in excess of 0.3% and the presence of synthetic cannabinoids are summarized in Table 1. Of the 25 products, 22 (88%) contained measurable quantities of CBD while 3 (12%) were devoid of phytocannabinoids. For those products containing CBD, total quantities per container varied greatly, with product R containing only 0.02 mg CBD, while product D contained 786 mg. Five products (20%) (H, J, R, X, Y) made no label claim for CBD, with product J and R containing 17 mg and 0.02 mg, respectively, while products H, X, and Y contained no measurable CBD. Interestingly, all five of these products were liquids for electronic cigarettes. In addition, five product labels provided no manufacturer's name or address, while nine others provided no physical address or only a web address.

For the 20 products that made a label claim for CBD, values for percent label claim ranged from as little as 0.06% (Product T) to as much as 168% (Product F). Only 3 products (12%) (A, K, U) were within ±20% of label claim for CBD, while 13 (52%) were <50% of claim.

Three products (A, E, N) had levels of THC that exceeded 0.3%, with 1 (Product N) being comprised of 45% THC. More disconcerting was the fact that 4 products (16%) (R, T, X, Y) were adulterated with synthetic cannabinoids. These are 4-fluoro MDMB-BUTINACA (methyl (S)-2-(1-(4-fluorobutyl)-1H-indazol-3-carboxamido)-3,3-dimethylbutanoate), 5-fluoro MDMB-PICA (N-[[1-(5-fluoropentyl)-1H-indol-3-yl]carbonyl]-3-methyl-L-valine, methyl ester), also known as 5-fluoro MDMB-2201 or just MDMB-2201, and R-5-fluoro ADB (N[[1-(5-fluoropentyl)-1H-indazol-3-yl]carbonyl]-3-methyl-D-valine, methyl ester) also known as 5-fluoro MDMB-PINACA. Three of these products were electronic cigarette vaping liquids.

In total, only 2 products (8%) (K, U) were within ±20% of label claim for CBD and not adulterated with either THC or synthetic cannabinoids.

Discussion

Since December 2018, a wide array of hemp-derived CBD products ranging in type from ingestible oils (e.g. neat oil, softgel capsules, gummies, beverages, etc.) to topical creams and inhalational (vaping) liquids have inundated the US market. Given the public's seemingly insatiable appetite for all things CBD and the confusion surrounding its regulatory status, the stage has been set for the introduction of products onto the marketplace with questionable quality. The results of this small pilot study confirm that suspicion. Based upon our findings it is clear that the labels of most CBD-containing products sampled do not accurately reflect the phytocannabinoid content present within the container. Our findings are in agreement with other recent surveys of CBD-containing products that found significant discrepancies in labeling accuracy (Bonn-Miller et al. 2017; Liebling et al. 2020). Over-labeling (products containing less than stated label claim) was the most common finding among CBD-containing products available in the state of Mississippi. From the consumer's perspective, and that of the Federal Trade Commission, over-labeling not only misrepresents the product but also signifies fraudulent business practices.

Table 1. Content versus label claim for CBD-containing products

Product	Product Name	Manufacturer	Product Type	CBD Label Claim	Total CBD	% Label Claim	THC > 0.3%	Synthetic Cannabinoids
A	Full Spectrum Hemp Oil	Functional Remedies (Louisville, CO)	ingestible oil	500 mg	521 mg	105%	+++	
B	Very Berry Syrup www.CreatingBetterDays.com	ingestible oil	200 mg	44 mg	22%			
C	CBD OIL	Green Roads (Davie, FL)	ingestible oil	7 mg	2.7 mg	39%		
D	CBD Full Spectrum (Gold)	Hemp Extracts (West Jordan, UT)	ingestible oil	500 mg	786 mg	157%		
E	Industrial Hemp Tincture	East Tennessee Hemp Company (no address)	ingestible hemp tincture	50 mg	38.5 mg	77%	+++	
F	EcoShot Relax Hemp	CBD Drip (Newport Beach, CA)	beverage "shot"	25 mg	42 mg	168%		
G	Max Chill (Fruit Punch)	Hemp Bombs (no address)	beverage "shot"	1.25 mg	0.2 mg	16%		
H	CBD Honey Sticks	no manufacturer's name, no address	CBD-containing honey	No claim	0 mg	???		
I	Prescott's Honey Sticks	Prescott's Industries (Burnsville, MS)	CBD-containing honey	25 mg	1.6 mg	6%		
J	SunMed CBD	Hemp Oil Products (Tampa, FL)	topical cream	No claim	17 mg	???		
K	Funky Farms Thin Mint	Arise Bioscience (Boca Raton, FL)	vape liquid	350 mg	417 mg	119%		
L	Zilla Pods CBD (Berry Wild Gelato)	Hempzilla LLC (www.hempzillacbd.com)	vape liquid	300 mg	0.3 mg	0.1%		
M	Royal CBD Classic	(www.royalcbd.com)	vape liquid	125 mg	10 mg	8%		
N	KR8ZY Glue Girl Scout Cookies	no manufacturer's name, no address	vape liquid	4-5 mg	0.5 mg	10%	+++ (45% THC)	
O	Exotic Watermelon Kush	(www.HempBombs.com)	vape liquid	75 mg	22 mg	30%		
P	Exotic Watermelon Kush		vape liquid	75 mg	17 mg	23%		

(*continued*)

Table 1. Continued.

Product	Product Name	Manufacturer	Product Type	CBD Label Claim	Total CBD	% Label Claim	THC > 0.3%	Synthetic Cannabinoids
		(www.HempBombs.com)						
Q	Silver Haze	VaporTech (Phoenix, AZ)	vape liquid	250 mg	134 mg	54%		
R	Blaze Vape	no manufacturer's name, no address	vape liquid	No claim	0.02 mg	???		+++ 4-fluoro MDMB
S	Exotic Watermelon Kush	(www.HempBombs.com)	vape liquid	4.5 mg	1.2 mg	27%		
T	Super Chill High Strength Cotton Candy	no manufacturer name, no address	vape liquid	1000 mg	0.6 mg	0.06%		+++ 5-fluoro MDMB
U	Blueberry Shatter	Blue Moon Hemp (no address)	vape liquid	99.6 mg	100 mg	100%		
V	Sweet Melons	Airbender Hemp (Uruguay)	vape liquid	200 mg	49 mg	24.5%		
W	Diamond CBD	Diamond CBD (no address)	vape liquid	500 mg	10 mg	2%		
X	Butterfly CBD Disposables (Blueberry)	no manufacturer's name, no address	vape liquid	No claim	0 mg	???		+++ 5-fluoro ADB
Y	Butterfly CBD Disposables (Strawberry)	no manufacturer's name, no address	vape liquid	No claim	0 mg	???		+++ 5-fluoro ADB

??? = indeterminate assessment
+++ = presence of THC >0.3% or synthetic cannabinoids

Despite anecdotal evidence that CBD has efficacy in a wide variety of ailments (e.g. chronic pain, epilepsy, sleep disorders, spasticity) only a small number of clinical trials have been conducted, and those utilized either Epidiolex® or other regulated products from outside the US (Pauli et al. 2020). Based upon our findings and those of others (Bonn-Miller et al. 2017; Liebling et al. 2020), generalizing the results of these trials to CBD-containing products widely available to the US consumer is ill-advised. While adverse effects have been recognized for CBD at doses higher than that recommended on most product labels found in this study (Huestis et al. 2019), the safety of chronic, low-dose CBD consumption remains to be determined. Thus, under-labeling, as was the case for 4 products in this study, may pose a risk for CBD overexposure with long-term usage. In the case of product N, THC was the under-labeled component, a finding that renders this product in violation of the Controlled Substances Act.

Perhaps the most alarming aspect of this study was the frequency with which CBD vaping products were adulterated with synthetic cannabinoids. Synthetic cannabinoids are a novel class of designer drugs intended to mimic the psychoactive effects of THC in marijuana (Pintori et al. 2017). They emerged as a new drug of abuse category in 2008. Synthetic cannabinoid use/abuse can have devastating toxicological consequences ranging from agitation and psychosis to arrhythmia, seizures, and death (Davidson et al. 2017; Armenian et al. 2018; Pacher et al. 2018). Their unpredictable toxicity may vary with dose, route of administration, individual vulnerability, and concomitant intake with other drugs (Davidson et al. 2017; Pintori et al. 2017). Compounding the harmful effects of synthetic cannabinoids when delivered *via* electronic cigarette is lung injury associated with vaping alone (Cherian et al. 2020). Recently, CBD oil vaping has been linked to a number of serious lung injuries (Conuel et al. 2019; Heinzerling et al. 2020). Our findings, while not conclusive, suggest that adulterated CBD products designed for e-cigarette use may exacerbate vaping-associated respiratory failure (Conuel et al. 2019; Fryman et al. 2020; Heinzerling et al. 2020). The number of liquid vaping products that provided no manufacturer information was also surprising as such practices only worsen the safety concerns surrounding these products.

In summary, labeling inaccuracies (over- or under-labeling) appear to be the norm for commercially available CBD products in the state of Mississippi. In addition, it is not uncommon for products to contain THC in excess of 0.3%. Finally, adulteration of CBD vaping products with synthetic cannabinoids poses a serious risk for adverse health effects. Taken together, our findings argue strongly for further development of current good manufacturing practices for CBD-containing products and their stringent enforcement.

Declaration of interest

The authors declare no conflicts of interest. The authors alone are responsible for the content and writing of the article.

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Bill J. Gurley is a Principal Scientist at the National Center for Natural Products Research at the University of Mississippi, Oxford. His research interests include mechanisms of herb-drug interactions, toxicity of multiple-component herbal dietary supplements, phytochemical modulation of

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Larry A. Walker is Director Emeritus of the National Center for Natural Products Research (NCNPR) at the University of Mississippi. He was trained in pharmacy and pharmacology, and has worked for much of his 38 yr career on natural products drug discovery, pharmacology, toxicology and metabolism. He has been integrally involved in the design and execution of preclinical and early clinical development studies on drugs and natural products.

Mahmoud A. ElSohly is a Research Professor at The National Center for Natural Products Research, and Professor of Pharmaceutics and Drug Delivery, School of Pharmacy, University of Mississippi (UM) and is the Director of the National Institute on Drug Abuse (NIDA) Marijuana Project at UM. He is also the President and Laboratory Director of ElSohly Laboratories Incorporated, an analytical forensic drug testing and product development laboratory. He received his undergraduate and Masters from Cairo University, Cairo, Egypt and his Ph.D. in 1975 from the University of Pittsburgh, School of Pharmacy, Pittsburgh, PA. He has been with the University of Mississippi since 1975 and has been Director of the NIDA Marijuana Project since 1981. He has over 40 years' experience working with the isolation of natural products (notably cannabis secondary metabolites), synthetic, analytical and forensic chemistry. He has more than 30 patents and over 300 publications in these areas of science. Dr. ElSohly is also a member of many professional organizations, such as American Society of Pharmacognosy, American Chemical Society, American Academy of Forensic Sciences, Society of Forensic Toxicology, International Cannabinoids Research Society, International Association of Cannabinoid Medicines, to name a few and have received numerous awards.

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